

FORMAL BUG REPORT

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SEVERITY: MODERATE | STATUS: OPEN

Issue ID	API-PERF-101	Date Logged	September 13, 2026
Component	Backend Routing (/api/products)	Environment	Localhost (Node.js/Express, MongoDB)
Test Phase	Automated API Validation	Reporting Tool	Postman Desktop App

Description

During the automated API regression testing phase of the `/api/products` endpoint, the server response time significantly exceeded the acceptable performance threshold of 500ms on the initial request (cold start), recording a latency of 1.20 seconds (1200ms).

Steps to Reproduce

- Initialize the local MERN backend server (Node.js) connected to the MongoDB Atlas cluster.
- Launch the Postman Desktop application.
- Execute a `GET` request to `http://localhost:5000/api/products`.
- Monitor the response time metrics in the network dashboard.

Expected Behavior

The API should successfully query the product catalog from the database and return a `200 OK` status with a JSON payload in **under 500ms**.

Actual Behavior:

The API returned the correct JSON array and a 200 OK status, but the transaction time was **1200ms**, causing the automated Postman performance assertion to fail.

Technical Context & Log Analysis

- **Failed Assertion:** `pm.expect(pm.response.responseTime).to.be.below(500);`
- Subsequent requests (warm state) drop to acceptable levels (< 200ms).
- **Root Cause Hypothesis:** The issue appears isolated to the initial MongoDB connection handshake overhead or lazy-loading resource allocation on the Express server.